

Low-Speed Turbulent Shear-Driven Mixing Layers with Large Thermal and Compositional Density Variations



Jon R. Baltzer and Daniel Livescu

Abstract Low speed shear-driven mixing layers involving fluid streams of different densities due to temperature or compositional variations are described by remarkably similar equations with some differences in the formulations of the molecular transport terms. These differences are related to specifics of the heat conduction and mass diffusion operators, as well as viscosity dependence on mixture molar mass and temperature in the low Mach number limit. Direct numerical simulations are performed in incompressible/low-speed limits to study the differences and similarities in mixing behavior associated with these configurations. The results demonstrate both subtle and significant changes in the mixing behavior for variable composition versus variable temperature mixing. Higher-order statistics related to density field reveal greater differences than are apparent from mean profiles; these differences can be extremely important when the physics is sensitive to mixing, such as in combustion problems. Therefore, conclusions regarding the mixing dynamics drawn from variable temperature mixing are not necessarily applicable to multi-species mixing.

Keywords Shear layers · Low speed · Variable density · Turbulence · Mixing

1 Introduction

Variable-density shear-driven mixing layers appear in a number of applications, including combustion. In combustion, the mixing of both different species (e.g. fuel and oxidizer) and temperature variations can influence the spatial variations in density. Shear-driven mixing involving only a single fluid with negligible thermal variations has historically received a great deal of attention both experimentally (e.g., Refs. [1–3]) and in numerical simulations (e.g., Refs. [4–8]). The single-fluid configuration contains much of the physics governing the variable-density shear layer, but

J. R. Baltzer · D. Livescu (✉)

Los Alamos National Laboratory, MS D413, Los Alamos, NM 87545, USA

e-mail: livescu@lanl.gov

J. R. Baltzer

e-mail: jbaltzer@lanl.gov

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significant differences have long been observed experimentally for variable-density mixing layers [9]. Local density variations exist as well in high-speed compressible mixing layers, which have also been studied extensively, particularly related to the strong reduction in mixing layer growth rate that occurs with increasing Mach number. Though density effects associated with compressibility were once thought to affect growth rate [9], direct numerical simulation (DNS) results have clarified how compressibility effects reduce the growth due to decreased turbulent kinetic energy production, as compressibility directly affects the pressure fluctuations, which decorrelates them from the strain (Refs. [10–13]). Recent simulations have further investigated the mixing characteristics of compressible mixing layers (e.g., Ref. [14]). However, low-speed mixing layers in which the two free streams have significantly different densities also have important applications, but have nonetheless been only scarcely studied. The low-speed configuration is the focus of the present study.

The pioneering 3D temporal simulations of Ref. [12] included an investigation of different free-stream densities within a broader study of compressible mixing layers. The differing densities were established by varying the temperature for a single fluid. They found that increasing Atwood number decreased the temporal thickness growth rate, though the extent depended on how thickness was defined. During self-similar growth, the Reynolds shear stress changed little in magnitude but its profile shifted to the light fluid side with increasing Atwood number. They also developed a model characterizing the shift of the mean velocity profile to the light fluid side and the associated decrease in momentum thickness growth rate. Mild compressibility effects were likely present because the convective Mach number was $M_c = 0.7$. More recently, Almagro et al. [15] performed DNS using a low-speed approximation for the flow of Ref. [12]. Two streams of a single fluid with different temperatures again create the density difference, but compressibility effects are considered negligible at low speeds. They also developed a semi-empirical model for the reduction in momentum thickness growth rate with density ratio.

Details of mixing layers with variable density due to differing fluid compositions are much less understood. Detailed studies of mixing layers involving two different miscible fluids have been rare, particularly when not complicated by other effects such as buoyancy or compressibility, despite earlier attention. The historic low-speed experiments of Ref. [9] using two gases with different densities found reductions in the growth rates as large as 50% for density ratios up to 7. These measurements were limited to mean density and streamwise velocity profiles and no details of the changes to turbulence and mixing properties are available. Our present investigation focuses on this flow but in a temporal configuration. Preliminary direct numerical simulations of mixing layers with miscible binary mixing at $A = 0.75$ are included in Ref. [16]; similar simulations using the same code and configuration are the basis for the present study.

The spatially-developing configuration, which is amenable to experimental studies, leads to thickening of the layer with downstream distance, while the temporal configuration, which is more convenient to address in numerical simulations, leads to a thickening of the layer in time at all streamwise locations. Though the symmetry associated with the periodic boundary conditions in the temporal configuration

changes the wave solutions relative to the spatially-developing set-up, there are also many similarities between the two flows, for example as related to the shear turbulence production.

2D simulations of early-time spatially-developing mixing layers show strong differences in entrainment depending on whether the low or high speed stream has lower or higher density [17]. Ashurts and Kerstein [18] studied variable density effects in temporal and spatial mixing layers using the one-dimensional turbulence stochastic simulation method; they captured many of the effects observed in Ref. [12].

Other studies have addressed variable-density shear-driven mixing layers with buoyancy or other complicating physics playing a significant role. Olson et al. [19] simulated mixing layers with mixed Rayleigh-Taylor (buoyant) and Kelvin-Helmholtz (shear) instability and Atwood numbers ranging up to 0.71 using the same governing as for our present study. Reference [20] simulated the mixing of vertical columns of fluid with different densities and perturbed interfaces. Gravity accelerates the perturbed heavy and light fluid columns in opposite directions within the triply-periodic domain to induce Kelvin-Helmholtz instability.

To understand the fundamental physics associated with density variations in a sheared flow, relevant mixing layers have been simulated with the same set-ups but using two different free-stream species with negligible thermodynamic variations [16] and using a single species with different temperatures of the two free streams [12, 15]. The present study briefly contrasts the governing equations, assumptions that lead to these equations, and mixing behavior based on DNS for these sets of governing equations. Effects of variations of species concentrations and temperature are also relevant to mixing layers appearing in combustion problems (e.g., Ref. [21]).

2 Governing Equations

The multicomponent compressible Navier-Stokes equations are applicable to all of the mixing layers considered here. Let partial derivatives be denoted by a subscript and the variable by which the partial derivative is taken follow a comma, with t representing the time variable and the index i representing the relevant spatial direction x_i . For a mixture of ideal gases in the low-Mach number limit, these equations become [22, 23]:

$$\rho_{,t} + (\rho u_j)_{,j} = 0, \quad (1)$$

$$(\rho u_i)_{,t} + (\rho u_i u_j)_{,j} = -\pi_{,i} + \tau_{ij,j}, \quad (2)$$

$$(\rho Y_\alpha)_{,t} + (\rho u_j Y_\alpha)_{,j} = -(\rho Y_\alpha V_{\alpha j})_{,j} + \dot{\omega}_\alpha, \quad (3)$$

$$\rho h_{,t} + \rho u_j h_{,j} = p_{0,t} - (\lambda T_{,j})_{,j} + \sum_\alpha (\rho Y_\alpha V_{\alpha j} h_\alpha)_{,j} + \Phi + \dot{Q}, \quad (4)$$

$$\begin{aligned}
u_{j,j} = & -\frac{1}{\gamma p_0} p_{0,t} + \frac{R}{c_p p_0} \left[-\sum_{\alpha} (\rho Y_{\alpha} V_{\alpha j}) h_{\alpha,j} + (\lambda T_{,j})_{,j} + \Phi + \dot{Q} \right] \\
& - \frac{1}{\rho R} \sum_{\alpha} (\rho Y_{\alpha} V_{\alpha j} R_{\alpha})_{,j} \\
& + \frac{1}{\rho} \left[\frac{\rho R}{c_p p_0} \dot{Q} + \sum_{\alpha} \left(\frac{R_{\alpha}}{R} - \frac{h_{\alpha}}{c_p T} \right) \dot{\omega}_{\alpha} \right], \tag{5}
\end{aligned}$$

where the viscous stress, assumed to be Newtonian, is

$$\tau_{ij} = \mu \left[u_{i,j} + u_{j,i} - \frac{2}{3} u_{k,k} \delta_{ij} \right]. \tag{6}$$

Equations (1) and (2) are the mass conservation and momentum equations governing the fields of density ρ , velocity components u_i , and dynamic pressure π . For the low-Mach number (M) approximation, the pressure p can be expanded in M and truncated to $p \approx p_0 + \pi$, where p_0 is thermodynamic pressure and π is the dynamic component [24]. Equation (3) governs the diffusion of species α , with Y_{α} representing each species' mass fraction. The diffusional velocity components are $V_{\alpha j}$. This system also includes the energy transport equation rewritten as a transport equation (Eq. 4) for enthalpy $h = e + p/\rho$, which is convenient for addressing a special case below [25]. Its transport is governed by the thermodynamic pressure change, thermal conduction, diffusion of species, and viscous dissipation Φ . The zeroth order pressure term p_0 can be shown to be spatially uniform but, in general, can vary with time [26]. Heat conduction is specified by Fourier's law with conduction coefficient λ . The diffusion velocities are given by the Maxwell relations [27]. For the binary case, they reduce to the Fickian form. Note that the Fickian and Fourier expressions of the mass diffusion and heat conduction terms are the low-Mach number limits of the full multicomponent operators. The low-Mach number limit leads to no further simplifications to the compressible form of the viscous stress tensor, as the divergence of velocity remains non-zero. A number of mechanisms can contribute to the divergence of velocity in Eq. (5). In general, non-zero divergence of velocity can be generated in the presence of background pressure changes, heat conduction, viscous dissipation, heat release, reactions between species with different molar masses, and mass diffusion. $\dot{Q}$ represents the rate of energy release from a heat source (such as from a chemical/nuclear reaction or radiation) and $\dot{\omega}_{\alpha}$ represents the rate of production associated with a scalar source for species α . The gas constant R for the ideal gas mixture is $R = \sum_{\alpha} R_{\alpha} Y_{\alpha}$.

For a mixture of ideal gases, the equations of state in the variable-density low-Mach number approximation is $p_0 = \rho R T$ and $h = \sum_{\alpha} Y_{\alpha} h_{\alpha}(T)$. Then the ideal gas relation simplifies to

$$\frac{1}{\rho} = \frac{T}{p_0} \sum_{\alpha} Y_{\alpha} R_{\alpha}, \tag{7}$$

where R_α and p_0 are constant. p_0 is approximated as spatially constant in many applications [26, 28], including mixing layers, and can also be shown to be temporally constant under certain circumstances described below.

Two special nonreacting cases that can sustain large density variations are here examined: (1) single species with variable temperature in the low-speed approximation and (2) two species with different molar masses in the incompressible limit.

2.1 Single-Fluid Case with Temperature Variations

The first case is based on the LMNOB (i.e. low-Mach number non-Oberbeck-Boussinesq) equations, which are recovered from the general low-Mach number equation set in Eqs. (1) and (5) by assuming a single species system. The LMNOB equations have been applied to a number of simulations including vertical convection in air [26, 29]. Similar sets of equations have also been applied to reacting flows [28, 30]. Though the evolution of p_0 is a consideration in more complex flows (e.g., Ref. [26]), in variable-density turbulent mixing layers it can be argued that p_0 remains constant in time because it is an open system [15].

If the viscous dissipation is neglected in Eq. (5), the velocity field divergence implied by the LMNOB equations with constant background pressure is

$$\nabla \cdot \mathbf{u} = \frac{1}{c_p} \nabla \cdot \left[\lambda \nabla \left(\frac{1}{\rho} \right) \right]. \quad (8)$$

The corresponding energy transport equation can be simplified from Eq. (4) for the single-fluid case as

$$(\rho T)_{,t} + (\rho u_i T)_{,i} = \frac{1}{c_p} (\lambda T_{,i})_{,i}, \quad (9)$$

after neglecting the viscous dissipation. This equation is redundant with the divergence relation of Eq. (10) when p_0 is constant. This further simplified set of equations is denoted herein as the low-Mach number thermal density variation (LMTV) equations.

Studies addressing the case of a single fluid with temperature variations often simplify the LMNOB (or in some instances their simplified LMTV version) equations by assuming constant values for material transport properties such as c_p , thermal conduction coefficient λ , and viscosity μ . The simulations herein are restricted to the simplified case of the LMTV equations in which c_p and λ are constant. For constant material properties, the velocity field divergence simplifies to

$$\nabla \cdot \mathbf{u} = \frac{\lambda}{c_p} \nabla^2 \left(\frac{1}{\rho} \right). \quad (10)$$

The complete set of equations that must be solved in this case are comprised of mass conservation Eq. (1), momentum balance Eq. (2), and velocity divergence condition Eq. (10). As the former two equations are evolved in a numerical simulation, the divergence can be enforced using a fractional step method that solves a Poisson equation for the pressure terms involving π . This is the approach used for the simulations shown below. Alternatively, rather than solving for the flow in primitive variables, the approach of Ref. [15] satisfies this system of equations but solves for the momentum vector Helmholtz-decomposed into divergence-free and curl-free components. This approach solves for the π pressure term through a Poisson equation.

Current applications of LMNOB equations (in their simplified LMTV form) to mixing layers assume temperature-independent thermal conductivity and viscosity, such as in the variable-density mixing layer simulations of Refs. [12, 15]. While the emphasis in such studies was simply to attain the desired density differences, it also should be considered how these assumptions relate to real fluid behavior. If both streams of the same ideal gas are at the same pressure, the density ratio between them is inversely proportional to the streams' temperature ratio. Thus, $A = 0.75$ requires a temperature ratio of 7 to attain the density ratio of 7. Gases can experience significant changes in thermal conductivities and specific heats as well as viscosities over these ranges of temperatures. For instance, for streams of air at 273.15 and 1912.05 K and atmospheric pressure producing a density ratio of 7, the hotter stream has 24% greater c_p , 354% greater λ and 251% greater μ than the colder stream. In the variable-density mixing layers that are the present focus, the non-Oberbeck-Boussinesq (NOB) effect of interest is related to the inertial effects of density variations. In many other applications, the most prominent NOB effects are related to transport property changes that are dependent on temperature. Many examples are related to Rayleigh-Bénard and other convection problems, such as vertical convection. Those NOB effects first arise when the Oberbeck-Boussinesq (OB) approximation of temperature-independent material parameters (kinematic viscosity, thermal diffusivity, heat conductivity, isobaric specific heat capacity c_p , and isobaric thermal expansion coefficient β) is violated (e.g., Ref. [31]). Additional considerations in LMNOB flows exist due to the change of density with temperature and pressure: in a closed volume, these effects can be balanced to conserve mass [26], but in mixing layers, and particularly with reactions releasing energy, the boundary conditions can be specified to allow expanding fluid to flow outward [32]. The low-speed variable-density mixing layer simulations of Ref. [15] incorporated boundary conditions based on this work.

2.2 Two-Species Variable-Density Incompressible Case

For multi-species mixtures, the incompressible limit is now considered. The incompressible limit is obtained by taking the infinite speed of sound c limit [33]. For a mixture of ideal gases, this limit requires that $T \rightarrow \infty$ because $c^2 = \gamma RT$. Furthermore, in this limit, $p_0 \rightarrow \infty$ simultaneously to maintain a finite ρ , according

to the ideal gas law equation of state Eq. (7) [25]. This limit eliminates density effects caused by thermal variations.

For non-reacting materials, the resulting expression for the velocity divergence is $\nabla \cdot \mathbf{u} = -\frac{1}{\rho R} \sum_{\alpha} (\rho Y_{\alpha} V_{\alpha j} R_{\alpha})_{,j}$. Maxwell's relations [27] are needed to solve for the diffusion velocities. For binary mixing [25, 33],

$$\nabla \cdot \mathbf{u} = -\nabla \cdot \left(\mathcal{D} \frac{\nabla \rho}{\rho} \right). \quad (11)$$

For binary mixing without any sources of species (i.e., reactions), Eq. (3) simplifies to

$$(\rho Y_{\alpha})_{,t} + (\rho Y_{\alpha} u_i)_{,i} = \mathcal{D} (\rho Y_{\alpha,i})_{,i}. \quad (12)$$

Defining the microdensities of the fluids as $\rho_{\alpha} = \frac{W_{\alpha}}{\mathcal{R}} \frac{p_0}{T}$, with W_{α} being the species molar mass and $\mathcal{R}$ the universal gas constant, the density of mixed fluid is related to species mass fractions as $\frac{1}{\rho} = \sum_{\alpha} \frac{Y_{\alpha}}{\rho_{\alpha}}$. For the incompressible limit considered above, it can be shown that $p_0/T = \rho R$ is constant in space and time, so the microdensities ρ_{α} are likewise constant [25]. For the binary mixing under consideration, the final relation between density and species mass fractions is

$$\frac{1}{\rho} = \frac{Y_1}{\rho_1} + \frac{Y_2}{\rho_2}, \quad (13)$$

where $Y_1 + Y_2 = 1$. More generally (i.e. for non-ideal gas equation of state), this mixing rule is valid when the microdensities ρ_1 and ρ_2 of the two fluids remain constant during mixing, such that the volume occupied by the mixture is equal to the volume occupied by the unmixed constituents at their respective microdensities. This condition is also satisfied by mixing liquids. Previously, the mixture relation of Eq. (13) was derived for two-liquid mixtures by Ref. [34] and used in the pioneering Rayleigh-Taylor simulation of Ref. [35].

For binary mixing, assuming that Eq. (13) holds, simply substituting Eq. (13) into Eq. (12) and combining with Eq. (1) also yields the divergence in Eq. (11). This derivation was given for two-fluid mixtures satisfying Eq. (13) by Ref. [36], while this same divergence was previously obtained by Ref. [34] for a two-liquid system. In contrast, the derivation described above began with the divergence given in Eq. (5) based on thermodynamic properties of the fluid and simplified to Eq. (11). The latter derivation based on the mass conservation produces the same divergence relation, which is a condition required for consistency [25, 33]. Thus, the species mass conservation equation is redundant with the divergence equation.

For the case of two-species variable-density incompressible mixing, mass conservation Eq. (1), momentum balance Eq. (2), and divergence relation Eq. (11) comprise a complete set of equations to describe the flow evolution. This may be referred to as the INBM (incompressible non-Boussinesq mixing) equations. In this application, the term “non-Boussinesq” is indicative of including the inertial effects associated

with large density fluctuations that would not be captured in the Boussinesq limit of two-species incompressible mixing. In this limit, the velocity divergence reduces to 0 and causes density to behave as a passive scalar in the absence of buoyancy [25].

Applications to Rayleigh-Taylor and related simulations (e.g., Refs. [19, 37, 38]) often assume that ν and $\mathcal{D}$ are constant. Under these assumptions, the velocity field divergence simplifies to

$$\nabla \cdot \mathbf{u} = -\mathcal{D}\nabla^2 (\ln \rho). \quad (14)$$

Computationally, the INBM equations can be solved using a variant of the fractional step method that advances Eqs. (2) and (1) while requiring the velocity field to satisfy the divergence Eq. (14) [37, 39]. This is the same procedure as the first method described for the LMTV equations, except with the different divergence that now implicitly causes Eqs. (12) and (13) to be satisfied. This technique has been applied to a number of Rayleigh-Taylor turbulence two-species mixing simulations (e.g., Ref. [37]) and has been further discussed in related references. It should be noted that in the present simulations, the governing equations are solved without the buoyant terms, whereas these terms play important roles in many of the traditional applications (e.g. Rayleigh-Taylor instability).

2.3 Discussion

Despite the differing diffusion operators, there are notable similarities in the forms of the equations for the INBM and LMTV cases. One of the species mass fractions in the former case plays an analogous role to temperature in the latter case. The transport equations for these quantities, Eqs. (12) and (9), have similar forms. It should be noted again that different simplifications lead to these equations. For instance, the LMTV equations can include a term representing the conversion of kinetic energy to thermal energy through viscous dissipation, a mechanism that has no analogy in INBM nonreacting species mixing. However, this effect is typically small and virtually always neglected in practice. The most pronounced difference between these equations pertains to their diffusion terms: the outer gradient includes a density product for Eq. (12), whereas density does not appear in the diffusion of Eq. (9). In both cases, there is an inversely proportional relationship between density and mass fraction/temperature (aside from an additive constant). The final velocity divergence forms are similar, but the divergence is proportional to the Laplacian of $-\ln \rho$ versus the Laplacian of $1/\rho$. As it was shown that substituting the density-mass fraction relation into the species mass transport equation and combining with the overall mass conservation equation can be used to obtain the INBM divergence, likewise substituting the equation of state into the energy transport equation and combining with the overall mass conservation equation leads to the LMTV divergence.

Besides this difference in the governing equations, assumptions about the fluid properties play important roles in the flow development. Assumptions related to the conduction and diffusion coefficients, λ and $\mathcal{D}$, as applicable to each case, have been

discussed above, but the appropriate prescription for fluid viscosity ν can also depend on the flow case. In the incompressible INBM variable density shear mixing layer simulations described in Ref. [16], the kinematic viscosity ν and diffusivity $\mathcal{D}$ are assumed constant, such that the Schmidt number $Sc = \nu/\mathcal{D}$ remains uniform with a value of 1. This same assumption was used in the Rayleigh-Taylor simulations discussed above. The choice of constant ν implies that dynamic viscosity $\mu \sim \rho$, whereas with real fluids there is typically a weaker dependence on density such as $\mu \sim \sqrt{\rho}$ [40]. For the INBM case, Chapman-Enskog theory predicts that if the two gases remain at constant temperature and pressure, $\mathcal{D}$ would remain constant independent of the concentration [41].

In the LMTV simulations of Ref. [15] addressing mixing layers involving a single fluid at different temperatures, the dynamic viscosity μ is instead assumed to be constant. Thus, the heat transfer analog of Schmidt number, the Prandtl number $Pr = \nu/\alpha = c_p\mu/\lambda$, is uniform (α is the thermal diffusivity $\lambda/(\rho c_p)$).

Another study in Ref. [20] performed an incompressible simulation of two vertical streams with different species set in motion by buoyancy. Viscosity μ was assumed to be uniform and constant but a constant Schmidt number of 1 was also maintained. Kinematic viscosity $\nu = \mu/\rho$ therefore spatially varied with density, and $Sc = 1$ necessitated that diffusivity also behaved similarly as $\mathcal{D} = \mu/\rho$. Substituting the density-dependent diffusivity into Eq. (11) results in a velocity field divergence of

$$\nabla \cdot \mathbf{u} = \mu \nabla^2 \left(\frac{1}{\rho} \right), \quad (15)$$

where μ is constant. Thus, the form of the equation is the same as that in Eq. (10) for the LMTV case with constant λ . This example illustrates that both the governing equations and assumptions on the diffusion behavior play equally significant roles in determining the divergence of the velocity field.

The objectives of the simulations are to compare the cases outlined above and to assess the differences associated with the mixing of the scalar quantity that affects the density (interpreted as species mass fraction in INBM or temperature in LMTV) for several forms of transport properties. These are constant versus proportional to $1/\rho$ diffusion coefficients for the scalar (if both are interpreted in the species transport framework) and proportional to ρ versus constant dynamic viscosity.

3 Simulation Set-Up

Direct numerical simulations are performed of temporally-growing planar shear-driven mixing layers. Significant variable-density effects are introduced by adopting an Atwood number of $A = 0.75$, which corresponds to a density ratio of 7 between the free streams. The simulation methodology is that described in Ref. [16] for the $A = 0.75$ variable density mixing layer, including the same simulation code. However, only the INBM case with ν and $\mathcal{D}$ constant was performed in that work,

whereas several additional cases are here performed: LMTV with ν constant and λ/c_p constant, LMTV with μ constant and λ/c_p constant, and also INBM with μ and $\mathcal{D}$ constant. The INBM and LMTV governing equations are simulated by enforcing the applicable velocity field divergence through a fractional step method as the continuity and momentum equations are evolved. The constant μ simulations are prescribed by $\mu = \rho_0 \nu_0$ based on the average density of the two streams, ρ_0 , and the same kinematic viscosity, ν_0 , that is uniform and constant in the constant ν simulations. Likewise, the constant values of λ/c_p of the LMTV simulations are, when divided by ρ_0 , equal to the constant value of $\mathcal{D}$ of the INBM simulations. Given the inverse/direct proportionality with density for some of the $\mathcal{D}$ and ν cases, the relatively large density ratio makes the effect of these transport properties significant.

For all of the present simulations, the domain size in the streamwise (x) and spanwise (z) directions is the same as in Ref. [16], while the cross-stream (y) dimension is thicker. The governing equations are supplemented by slip wall boundary conditions in the y direction and periodic boundary conditions in the x and z directions. The simulations use a resolution of $2048 \times 2048 \times 512$ grid points. Each simulation is initialized by a thin interface prescribed by aligned $\tanh$ profiles of mean streamwise velocity and density to maintain the required velocity difference ΔU and density difference $\Delta \rho$ between the two free streams. ΔU , $\Delta \rho$, and the average of the two free-stream densities, ρ_0 , remain the same for each simulation. The initial velocity profile is perturbed to initiate the transition to turbulence (as described in Ref. [16]). Each simulation begins from the same initial conditions to minimize the statistical influence of the initial disturbance when comparing the differences produced by differing governing equations.

4 Results

The primary objective of the simulations is to determine how the flow evolution according to the different governing equations described above affects the most basic flow properties and mixing. This is evaluated after the flow has had time to develop and the mixing layer is growing in an approximately self-similar manner. Based on a number of indications, such as growth rate and peak Reynolds stress time histories, the time around $t \Delta U / h_0 = 284$ displays the best adherence to self-similar growth behavior. h_0 is the initial thickness of the mixing layer streamwise velocity profile. The Reynolds numbers are approximately $Re_\lambda = \tilde{k} \sqrt{20/(3\nu\varepsilon)} = 73$ during this time period, where $\tilde{k}$ is turbulent kinetic energy and ε is its dissipation rate. The statistical profiles (e.g., mean streamwise velocity and mean density) remain constant when y is scaled by the time-growing thickness h . h is defined to be the distance in y between the points at which the mean streamwise velocity $\tilde{U}_1$ is at 10% and 90% of the free-stream velocity difference ΔU . Momentum thickness $\delta_m = \int (\bar{\rho}/\rho_0)(1/4 - \tilde{U}^2/\Delta U^2) dy$ may alternatively be used to measure the mixing layer thickness; during self-similar growth, $h \approx 7.9\delta_m$ in the flows shown, but the relation varies with Atwood number and is not valid at early times. $\bar{f}$ denotes ensemble averaging of quantity f ; in the

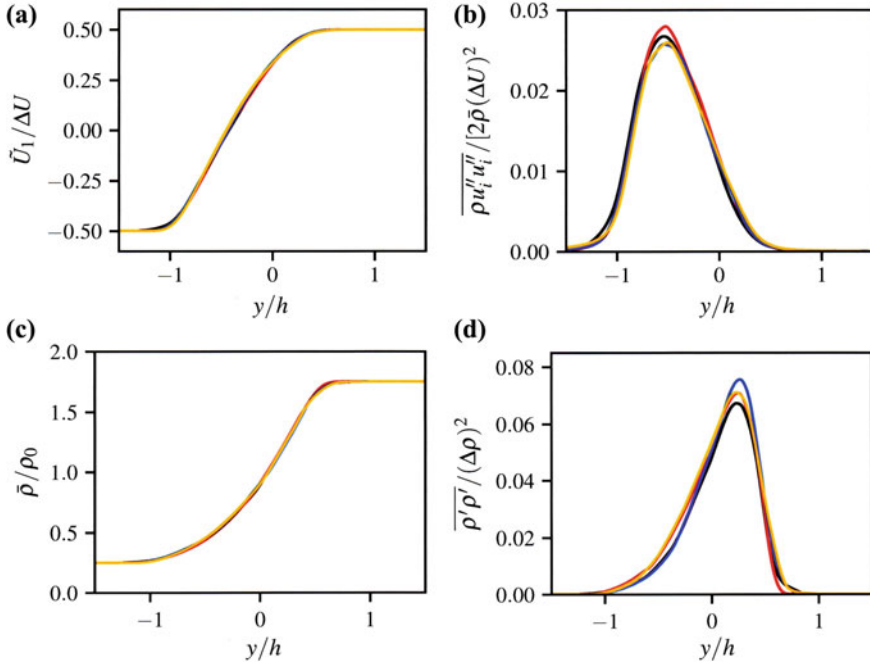


Fig. 1 Comparison of self-similar growth **a** mean streamwise velocity, **b** turbulent kinetic energy, **c** mean density, and **d** density variance profiles for (—) $\mathcal{D} = \text{const.}$ (INBM) and $\mu \propto \rho$, (—) $\mathcal{D} = \text{const.}$ and $\mu = \text{const.}$, (—) $\mathcal{D} \propto 1/\rho$ (constant-transport-property LMTV) and $\mu \propto \rho$, and (—) $\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$

simulation data, it is approximated by area averaging over the homogeneous x and z coordinates. Self-similar averages are in addition averaged over the appropriate time period with y scaled by h . The mean streamwise velocity is Favre averaged as $\tilde{U}_1 = \overline{\rho u_1} / \bar{\rho}$. Fluctuations are defined relative to Reynolds averages as $f' = f - \bar{f}$ and relative to Favre averages as $f'' = f - \tilde{f}$. Statistical profiles are averaged between $t \Delta U / h_0 = 227$ and 341 to reduce statistical noise; profiles for mean density and streamwise velocity are shown in Fig. 1.

Overall, the most intense turbulence moves to the light-fluid (negative y) side, which causes the gradient in mean streamwise velocity to thicken more quickly to the light-fluid side. The turbulence intensity is quantified by turbulent kinetic energy. The associated position of strongest turbulent mixing also makes the mean density profile gradient shallower on the light fluid side and steeper on the heavy fluid side. The mean profiles for both density and streamwise velocity are nearly identical between INBM versus LMTV cases and constant ν versus μ cases. However, turbulent kinetic energy and density variance profiles weakly vary between cases, indicating some degree of mixing differences.

Table 1 y/h values for neutral points and peak values of density and velocity with $A = 0.75$

Equation Form	η_ρ ($\bar{\rho} = \rho_0$)	η_2 ($\tilde{U}_2$ peak)	η_1 ($\tilde{U}_1 = 0$)	η_{12} ($\tilde{R}_{12}$ peak)
$\mathcal{D} = \text{const. (INBM) and } \mu \propto \rho$	0.07	-0.36	-0.43	-0.55
$\mathcal{D} = \text{const. and } \mu = \text{const.}$	0.07	-0.27	-0.44	-0.54
$\mathcal{D} \propto 1/\rho$ (LMTV) and $\mu \propto \rho$	0.08	-0.31	-0.44	-0.54
$\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$	0.07	-0.37	-0.45	-0.53

For the same constant- $\mathcal{D}$ INBM and constant- λ and c_p LMTV cases shown in Fig. 1, the y/h values of the neutral points and peak values of various quantities are shown in Table 1. The points of interest include the neutral point of density (where the mean density is equal to the average of the free streams), the peak value of cross-stream velocity (the negative $\tilde{U}_1$ near the interface only is present in variable-density temporal mixing layers), the neutral point of streamwise velocity (where $\tilde{U}_1$ is equal to 0, the average of the free stream values), and the peak value of $\tilde{R}_{12}$. Reynolds stress $\tilde{R}_{12} = \overline{\rho u'_1 u'_2} / \bar{\rho}$ is an important quantity closely related to the growth of the mixing layer [10, 12]. The positions remain consistent among cases. The cross-stream mean velocity $\tilde{U}_2$ peak position varies the most, but statistical variation is likely more pronounced for this quantity as the $\tilde{U}_2$ component is relatively weak. In each case, $\eta_2 > \eta_1, \eta_\rho$, the position of neutral density, changes the least from the initial interface $y = 0$ and moves slightly to the heavy fluid side. The streamwise velocity neutral points move significantly to the light fluid side and represent the drifts of each mixing layer, while the $\tilde{R}_{12}$ peaks associated with intense turbulence are consistently furthest to the light fluid side.

The density fluctuation mean squared statistics shown in Fig. 1 provide further information on the mixing. Besides the weak differences in peak magnitudes, the tails on the heavy fluid side decay more gradually for the constant ν cases. As it is shown below from density probability density functions (PDFs), stirring parcels of mostly-unmixed heavy fluid cause larger density fluctuations on the heavy fluid side, while more mixed fluid on the lighter-fluid side is associated with weaker fluctuation intensity.

For binary species mixing, analogies with reacting flows (e.g., combustion) suggest forming statistics based on the mass fraction of the heavy fluid. Based on Eq. (13) for the INBM case, this mass fraction is related to specific volume $v = 1/\rho$ as

$$Y_2 = \frac{-\rho_1 \rho_2}{\rho_2 - \rho_1} v + \frac{\rho_2}{\rho_2 - \rho_1}. \quad (16)$$

It follows that the means and variances can be written as $\bar{Y}_2 = -[\rho_1 \rho_2 / (\rho_2 - \rho_1)] \bar{v} + \rho_2 / (\rho_2 - \rho_1)$ and $Y_2'^2 = [\rho_1 \rho_2 / (\rho_2 - \rho_1)]^2 \bar{v}^2$. In the low Atwood number limit, the heavy fluid mass fraction plays a role approaching that of a passive scalar. For this reason, a scalar designated by ϕ will be used to represent Y_2 , with the understanding

that this scalar affects variable-density flow dynamics. The analogy between INBM mass fraction and LMTV temperature suggests defining the scalar for the LMTV case as $\phi = (T - T_1)/(T_2 - T_1)$. Thus, ϕ also varies from 0 to 1 between the low density (high T) and high density (low T) fluid streams. Substituting T obtained from Eq. (7) into the above expression yields $\phi = (v - v_1)/(v_2 - v_1)$. Substituting $v_1 = 1/\rho_1$ and $v_2 = 1/\rho_2$ yields the same right-hand side as Eq. (16). Thus, this ϕ based on scaled temperature for the LMTV case obeys the same relation with specific volume as the incompressible INBM case.

These equivalencies suggest comparing statistics of the scalar between cases. The specific volume variance is normalized by dividing by $(v_1 - v_2)^2$, the difference in free-stream specific volumes squared; this is equivalent to the $[\rho_1 \rho_2 / (\rho_2 - \rho_1)]^2$ multiplicative factor above. The scalar statistics are shown in Fig. 2. Unlike the density variance, the scalar (and thus specific volume) variance peak is concentrated in the light fluid side and exhibits large differences among cases, which is indicative of variable density effects. The constant- μ simulation peak values are consistently greater than those of the constant- ν peaks, while the constant- $\mathcal{D}$ (INBM) peak is larger than

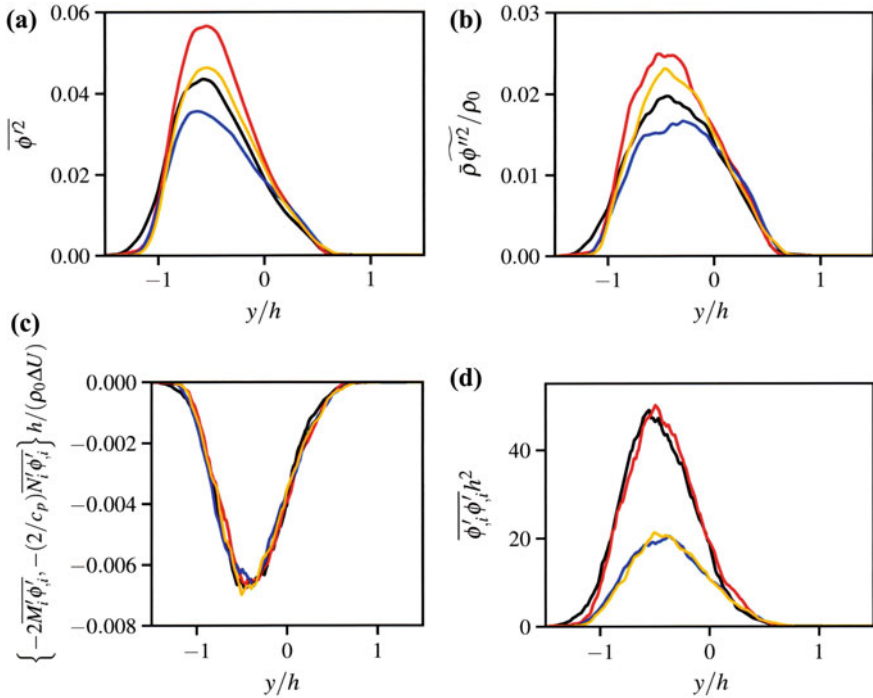


Fig. 2 Comparison of **a** self-similar growth scalar ϕ (or scaled specific volume) variance profiles, as well as $t \Delta U / h_0 = 284$, **b** density-multiplied Favre-averaged scalar variance, **c** scalar dissipation, and **d** scalar gradient intensity profiles. Colors are (—) $\mathcal{D} = \text{const.}$ (INBM) and $\mu \propto \rho$, (—) $\mathcal{D} = \text{const.}$ and $\mu = \text{const.}$, (—) $\mathcal{D} \propto 1/\rho$ (constant-transport-property LMTV) and $\mu \propto \rho$, and (—) $\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$

the $\mathcal{D} \propto 1/\rho$ (LMTV) peak for a given viscosity prescription. The Favre scalar variance peak also moves to the light fluid side, however the cases with similar viscosity prescriptions yield closer magnitudes. Thus, the scalar variance is more sensitive to the viscosity variation than diffusion variation, though both of these effects influence the results, for the Reynolds number examined here. Given the linear relationship between specific volume and temperature, the variances for specific volume can also be compared with those for temperature reported in Ref. [15], which corresponds to the LMTV constant- μ case. The results are fully consistent with those shown here.

To clarify the factors that affect the scalar variance, its budget equation is frequently investigated. Starting from species conservation in Eq. (12), the budget equation for ϕ variance in the constant- $\mathcal{D}$ INBM cases can be expressed in the following form [42]

$$\begin{aligned} \left(\bar{\rho} \widetilde{\phi'^2} \right)_{,t} = & \underbrace{- \left(\bar{\rho} \tilde{U}_i \widetilde{\phi'^2} \right)_{,i}}_{\text{term I}} \underbrace{- 2 \bar{\rho} \widetilde{u''_i \phi' \tilde{\phi}}_{,i}}_{\text{term II}} \underbrace{- \frac{2 \bar{M}'_i \phi'}{c_p}}_{\text{term III}} - \underbrace{\left(\bar{\rho} \widetilde{u''_i \phi'^2} \right)_{,i}}_{\text{term IV}} \\ & + 2 \underbrace{\left(\bar{M}'_i \phi' \right)_{,i}}_{\text{term V}} + \underbrace{2 \bar{\phi''} \bar{M}_{i,i}}_{\text{term VI}}, \end{aligned} \quad (17)$$

where $M_i = \rho \mathcal{D} \phi_{,i}$. The terms contributing to the scalar variance evolution are (I) convection by mean velocity, (II) production, (III) dissipation, (IV) turbulent diffusion, (V) molecular diffusion, and (VI) compressibility. The scalar dissipation term is of particular interest because it represents the rate of mixing and also is directly related to reaction rate in simplified combustion problems with fast chemistry [43].

For LMTV cases, the scalar ϕ defined above as scaled temperature obeys a similar budget equation for its Favre variance. For constant c_p , this equation is:

$$\begin{aligned} \left(\bar{\rho} \widetilde{\phi'^2} \right)_{,t} = & \underbrace{- \left(\bar{\rho} \tilde{U}_i \widetilde{\phi'^2} \right)_{,i}}_{\text{term I}} \underbrace{- 2 \bar{\rho} \widetilde{u''_i \phi' \tilde{\phi}}_{,i}}_{\text{term II}} \underbrace{- \frac{2 \bar{N}'_i \phi'}{c_p}}_{\text{term III}} - \underbrace{\left(\bar{\rho} \widetilde{u''_i \phi'^2} \right)_{,i}}_{\text{term IV}} \\ & + \underbrace{\frac{2}{c_p} \left(\bar{N}'_i \phi' \right)_{,i}}_{\text{term V}} + \underbrace{\frac{2}{c_p} \bar{\phi''} \bar{N}_{i,i}}_{\text{term VI}}, \end{aligned} \quad (18)$$

where $N_i = \lambda \phi_{,i}$. The budget terms have similar interpretations as above. The most important difference in the form of the budget equations between the INBM and LMTV cases is that M includes density but N does not.

Unlike the Favre-averaged scalar variance itself, the corresponding dissipation terms have very similar behavior, both in magnitude and spatial distribution, for all of the cases (Fig. 2). When the diffusion coefficient ($\mathcal{D}$ or λ) is constant, the scalar dissipation term is essentially the variance of the scalar gradient. When written for the constant- $\mathcal{D}$ INBM governing equations, it is weighted by density (in M_i), but that weighting is absent for the corresponding constant- c_p and λ LMTV case (in N_i).

Examining the scalar gradient fluctuation intensity directly (Fig. 2), without density weighting for any case, demonstrates a strong dependence in magnitude on the form of the diffusivity. The inertially-induced drift of the mean velocity profile relative to the mean density profile suggests that fluid in the region of strongest turbulence and mixing experiences larger diffusivity for constant λ LMTV relative to constant $\mathcal{D}$ INBM. This is a consequence of the mixing layers growing preferentially into the lighter stream, though the strongest shear and disturbance were initially positioned at the fluid interface. Since $\mathcal{D} \propto 1/\rho$ is equivalent to constant λ , this drifting implies that $\mathcal{D}$ is larger than average (i.e., the average of the two free streams) in this region. The diffusivity of the constant $\mathcal{D}$ simulations is only matched by the $\mathcal{D} \propto 1/\rho$ simulations where $\rho = \rho_0 = (\rho_1 + \rho_2)/2$. Since the strongest turbulence and mixing becomes centered where the local density is, on average, less than ρ_0 , it follows that the local $\mathcal{D}$ is larger than that of a constant $\mathcal{D}$ simulation. Despite the differences in the ϕ gradients, however, the weighting by density results in similar magnitudes of scalar dissipation for the INBM and LMTV cases. Similar arguments can be made that, in the same region, the local viscosity ν of a constant μ simulation is somewhat larger than that of a constant ν simulation. However, the scalar gradients display less sensitivity to viscosity. The effects on mixing layers of streams having differing viscosities (but the same densities) has been studied in depth (e.g., Ref. [44]). The comparisons made in Fig. 2 indicate that the differing transport forms modify the flow structure, but a number of gross quantities (e.g., scalar dissipation, as well as the mean density and velocity profiles) are only weakly affected. In summary, in these simulations, the scalar variance is relatively sensitive to the forms of the scalar diffusion and fluid viscosity, but its dissipation is insensitive to both forms.

The differences in statistics involving higher-order quantities are indicative of differing mixing dynamics. The PDFs of the density shown in Fig. 3 for selected y positions further illustrate the differences. Since local density is related to specific volume as $v = 1/\rho$ and specific volume has a linear relationship to species mass fraction or temperature depending on the case, these plots can also be interpreted in terms of the latter quantities. While the distributions are similar among cases at each y location, details of the PDFs reveal significant differences. At $y/h = -0.47$, peaks for both of the $\mathcal{D} \propto 1/\rho$ (LMTV) cases are located at densities greater than that of the pure light fluid, indicating that mixed fluid dominates. Conversely, for each of the constant $\mathcal{D}$ (INBM) cases (black and red curves), a peak also occurs at the lightest fluid density ($\rho/\rho_0 = 0.25$); this tail indicates that measurable traces of light fluid have yet to begin mixing. As y increases slightly, the behavior that dominates the intensely mixed layer emerges: a single dominant PDF peak with position weakly biased toward heavier density at constant ν than constant μ but slightly lower probability of fluid at higher density, indicating that the heavier fluid mixes more for constant ν . In addition, there are lower probabilities of fluid at the lowest densities, indicating the lightest fluid is also more mixed. As y further increases, each constant $\mathcal{D} \propto 1/\rho$ (LMTV) PDF develops a tail of increasing value near the pure heavy fluid. This effect continues as y position increases, with the unmixed heavy fluid peak much larger for the $\mathcal{D} \propto 1/\rho$ cases at the highest y value shown.

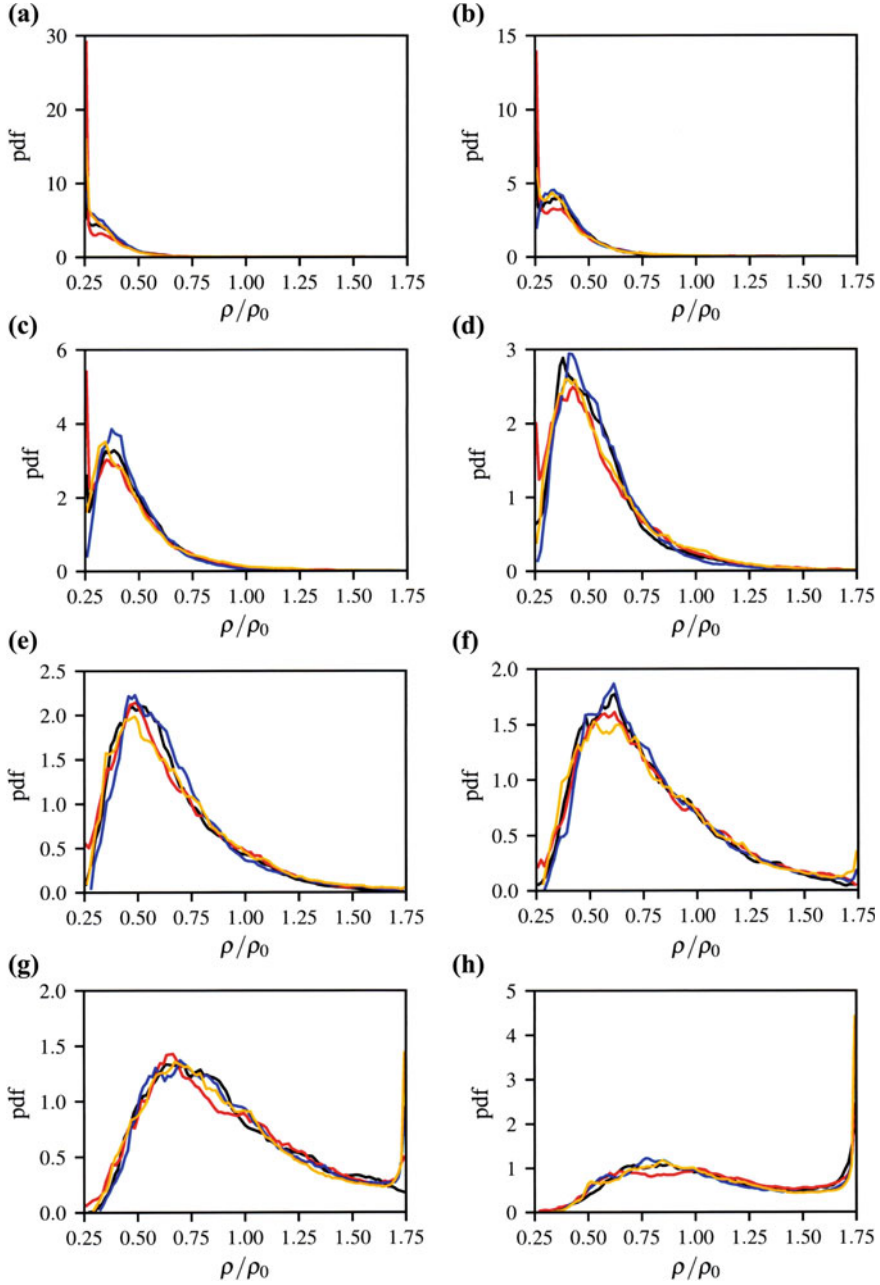


Fig. 3 Comparison of density PDFs at $t\Delta U/h_0 = 284$ and $y/h =$ **a** -0.70 , **b** -0.58 , **c** -0.47 , **d** -0.35 , **e** -0.23 , **f** -0.12 , **g** 0.00 , and **h** 0.12 . (—) $\mathcal{D} = \text{const.}$ (INBM) and $\mu \propto \rho$, (—) $\mathcal{D} = \text{const.}$ and $\mu = \text{const.}$, (—) $\mathcal{D} \propto 1/\rho$ (constant-transport-property LMTV) and $\mu \propto \rho$, and (—) $\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$

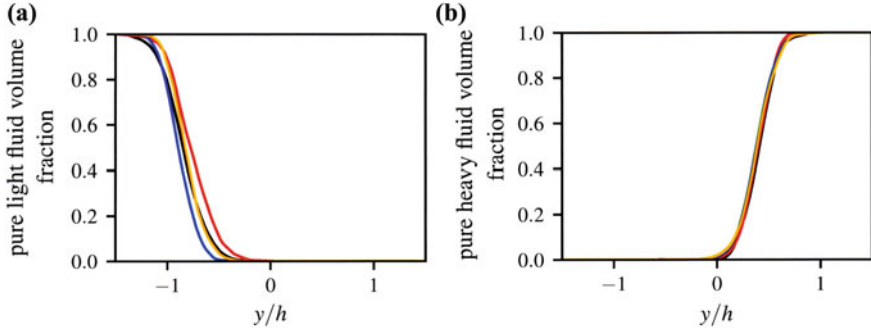
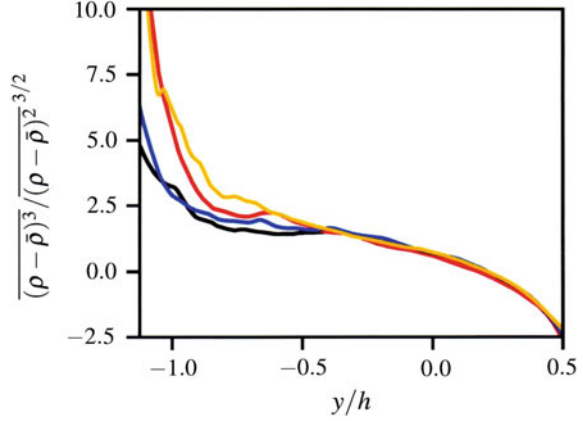


Fig. 4 Comparison of volume fractions of pure **a** light and **b** heavy fluid (defined as density within 1% of the total difference from each pure value) at $t\Delta U/h_0 = 284$. Colors are (—) $\mathcal{D} = \text{const.}$ (INBM) and $\mu \propto \rho$, (—) $\mathcal{D} = \text{const.}$ and $\mu = \text{const.}$, (—) $\mathcal{D} \propto 1/\rho$ (constant-transport-property LMTV) and $\mu \propto \rho$, and (—) $\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$

The differences in the PDF tails can be compared by plotting the volume fractions of pure fluid as functions of y (Fig. 4). These comparisons show that the mixing behaviors involving the pure light sides are more significantly modified among the different cases than those involving the pure heavy fluid. On the light fluid side, the edge of the layer (where the pure light fluid fraction begins to decay from 1.0) is positioned at more strongly negative y values for the constant- ν cases than for the constant- μ cases. These extents are generally insensitive to the form of diffusivity. However, the PDFs include complicated peak behavior at positions where pure light fluid is beginning to mix. The 1% threshold used here does not correspond directly to the PDFs, which include bins of finer granularity that reveal features such as the weak peak of unmixed light fluid at $y/h = -0.47$ for the constant- ν INBM case. From each outer edge, Fig. 4 shows that the pure light fluid persists further into the mixing layer for the constant $\mathcal{D}$ cases than for the $\mathcal{D} \propto 1/\rho$ cases. In other words, the pure light fluid mixes more rapidly with respect to position towards the center of the layer with $\mathcal{D} \propto 1/\rho$. While caution must be observed in interpreting the curves as functions of the scaled coordinates y/h with the possibility of altered profile shapes and growth dynamics affecting how the curves lie in relation to each other, the growths and mean profiles are very similar between cases, which supports the validity of these comparisons. The behaviors of the pure light fluid curves suggest that the outer boundary of mixing is mainly controlled by the viscosity, but as the layer is entered, the scalar diffusivity controls how quickly mixing eliminates pure light fluid.

In regard to the pure heavy fluid side, mixing begins at (that is, only pure heavy fluid exists up to) similar points for all of the cases. Some amount of pure heavy fluid persists for slightly further distances into the mixing layer for the LMTV (or $\mathcal{D} \propto 1/\rho$) cases. Thus, diffusivity also affects how quickly the mixing eliminates pure heavy fluid.

Fig. 5 Comparison of density skewness at $t\Delta U/h_0 = 284$. The y range is restricted to the interior of the layer where turbulent motions are most active. Colors are (—) $\mathcal{D} = \text{const.}$ (INBM) and $\mu \propto \rho$, (—) $\mu = \text{const.}$, and $\mathcal{D} = \text{const.}$, (—) $\mathcal{D} \propto 1/\rho$ (constant-transport-property LMTV) and $\mu \propto \rho$, and (—) $\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$



The asymmetries of the density PDFs inside the layer can be quantified by their skewnesses. These are plotted across the layers in Fig. 5. All of the cases collapse near the heavy fluid side as the skewnesses becomes negative. This collapse occurs despite the significantly different magnitudes among cases of the squared density fluctuation $\overline{\rho'\rho'}$ in the tail on the heavy-fluid side (Fig. 1). Approaching the light-fluid side, the skewnesses behave similarly but diverge among cases. In general, the constant μ cases are more positively skewed as the light-fluid edge of the mixing layers is approached. This is consistent with their pure light fluid tails persisting further into the mixing layer and beginning to decay at smaller distances from the core.

The complex dependencies on diffusivity and viscosity suggest that the various scales of motion are affected differently. Spectra of density and scalar ϕ further clarify these effects. Figure 6 shows two-dimensional spectra of density for various cross-stream (y) positions. Though the flow is anisotropic between the x and z axes due to the mean shear, averaging implied by the two-dimensional spectra is sufficient to reveal the scale dependence. Particularly at the lowest y values, the low wavenumbers of the density field are more strongly affected by the fluid viscosity form, with the diffusivity form having little effect. At the highest wavenumbers, particularly around the location of the most intense turbulence (e.g., at $y/h = -0.23$), the spectra are strongly affected by the diffusivities (i.e., INBM with constant $\mathcal{D}$ versus LMTV with constant λ and c_p). This diverging behavior of spectra at high wavenumbers with respect to diffusivity is even more pronounced when the scalar is considered rather than the density, and this effect persists over all y positions shown (Fig. 7).

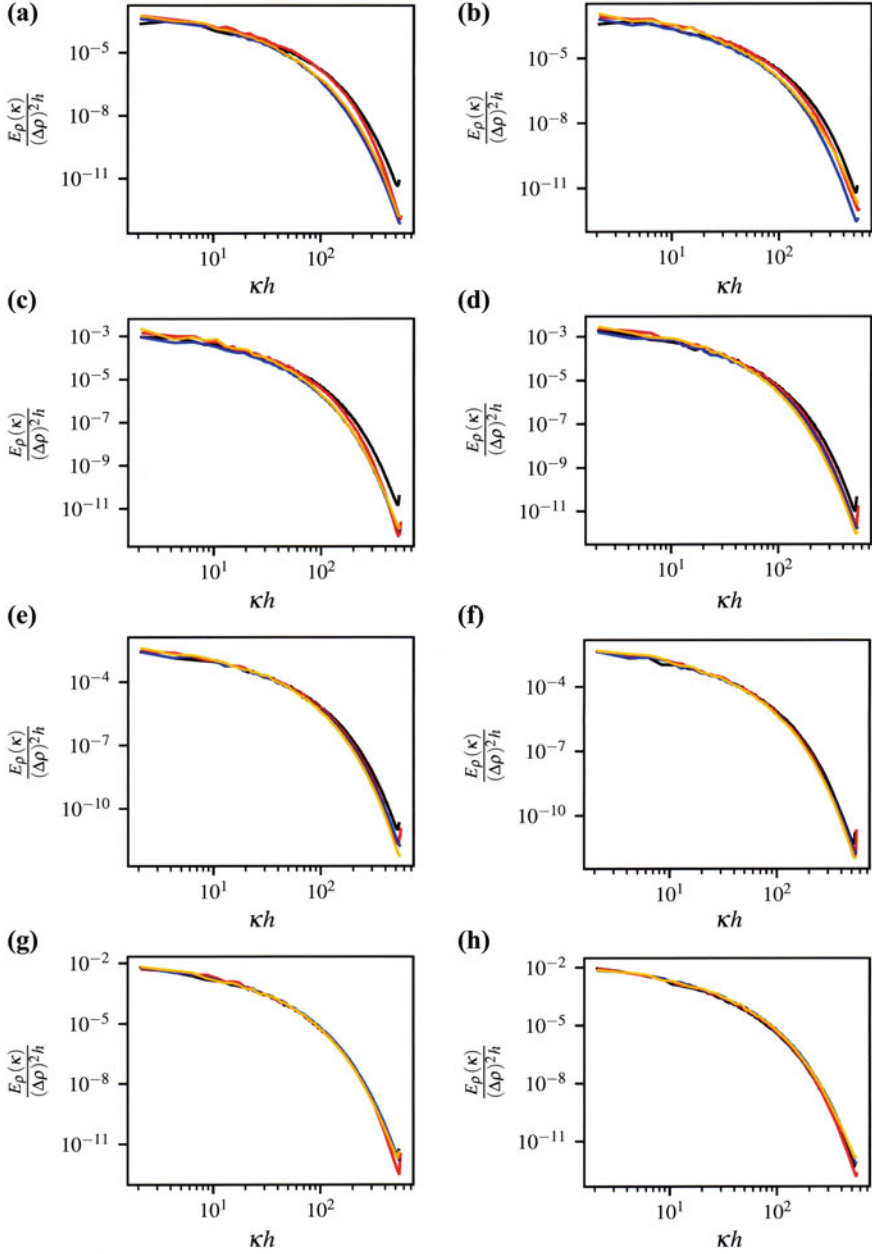


Fig. 6 Comparison of 2D spectra for density at $t\Delta U/h_0 = 284$ and $y/h =$ **a** -0.70 , **b** -0.58 , **c** -0.47 , **d** -0.35 , **e** -0.23 , **f** -0.12 , **g** 0.00 , and **h** 0.12 . (—) $\mathcal{D} = \text{const.}$ (INBM) and $\mu \propto \rho$, (—) $\mathcal{D} = \text{const.}$ and $\mu = \text{const.}$, (—) $\mathcal{D} \propto 1/\rho$ (constant-transport-property LMTV) and $\mu \propto \rho$, and (—) $\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$

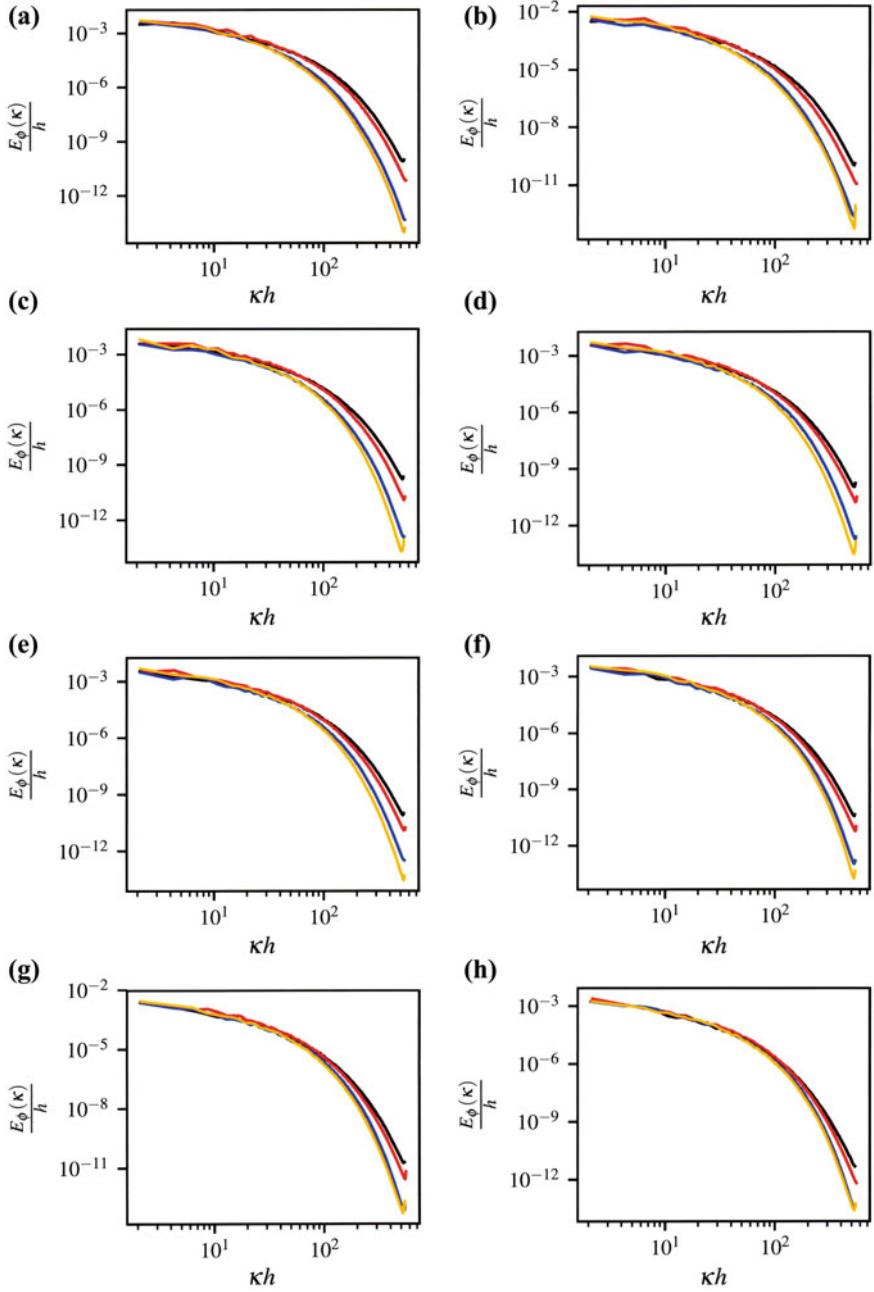


Fig. 7 Comparison of 2D spectra for scalar ϕ (heavy fluid mass fraction or scaled temperature) at $t\Delta U/h_0 = 284$ and $y/h =$ **a** -0.70 , **b** -0.58 , **c** -0.47 , **d** -0.35 , **e** -0.23 , **f** -0.12 , **g** 0.00 , and **h** 0.12 . (—) $\mathcal{D} = \text{const.}$ (INBM) and $\mu \propto \rho$, (—) $\mathcal{D} = \text{const.}$ and $\mu = \text{const.}$, (—) $\mathcal{D} \propto 1/\rho$ (constant-transport-property LMTV) and $\mu \propto \rho$, and (—) $\mathcal{D} \propto 1/\rho$ and $\mu = \text{const.}$

5 Conclusions

Low- and zero-Mach number approximations have been employed for studying flows with large density variations in a variety of flows. In these limits, density variations can be related to two classes of flows: (a) single fluid flows with temperature fluctuations and (b) mixing between species with different densities (e.g. due to different molar masses). In addition, the associated changes in the fluid properties can also be important.

For temporal shear driven mixing layers, despite the similarities in the mean flow quantities, higher order statistics reveal significant differences in the mixing dynamics resulting from governing equations derived for density variations induced by species or thermodynamic variations. These changes are also indicative of modified turbulence structures. While the results shown are for a single Atwood number and the effects can be expected to increase in magnitude with Atwood number, they demonstrate that in general care must be taken to apply conclusions between dissimilar diffusion operators (e.g., LMTV or INBM) and other transport properties. In addition, very different physical conditions and simplifications (e.g., assuming that c_p and λ are constant despite large temperature variations) are necessary to reach the respective forms of governing equations for INBM (incompressible non-Boussinesq mixing) and LMTV (low-Mach number temperature variations), as they have been used in previous studies. Thus, though both the INBM and LMTV governing equations can be derived from the low-Mach number non-Oberbeck-Boussinesq equations for a multi-fluid mixture, distinct simplifications are required that have unique physical interpretations. Beyond the simplification to a single fluid or taking the infinite speed-of-sound incompressible limit, further assumptions on the transport properties (diffusivity or conduction coefficient) and their dependencies on density or temperature influence the flow evolution. These assumptions have differing fidelities relative to practical flow conditions. Physically obtaining high Atwood numbers in a LMTV configuration would imply large differences in temperature, which would result in large variations in fluid properties. Though these variations could be simulated as well, considering them weakens the connection to INBM flows and suggests that care is necessary in making analogies between practical flows. While the INBM equations solved herein assume a constant diffusivity, that assumption more closely matches the conditions experienced within a practical experiment. The results demonstrate that the differences between the LMTV or INBM governing equations, despite their close resemblance, can nevertheless significantly affect the mixing dynamics to which reacting flows are sensitive. Exploring the range of species- and temperature-induced density effects is thus necessary to gain the physical insight necessary to advance the prediction of the broad spectrum of flows in which mixing plays a critical role.

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